End-to-End Aspect-Based Sentiment Analysis with Cascaded ATE and ASC on SemEval-2014

Yubo Wang Gousu Ding Fangyuan Zhang Yunzhu Chen

Abstract

Aspect-based sentiment analysis (ABSA) in real review settings requires both locating opinion targets (aspect terms) and assigning polarity to each target. We study a practical cascade in which a span-level aspect term extraction (ATE) model feeds an aspect sentiment classification (ASC) model that consumes sentence–aspect pairs. Using the SemEval-2014 Task 4 restaurant and laptop benchmarks (Pontiki et al., 2014), we train domain-specific ATE and ASC modules with two encoder backbones, BERT-base (Devlin et al., 2019) and DeBERTa-base (He et al., 2021), and evaluate the full pipeline under in-domain and cross-domain test conditions. We report ATE precision, recall, and F1, along with ASC accuracy and macro-F1 under gold versus predicted aspects, where the latter reflects true deployment. Quantitatively, ATE F1 collapses under domain shift, while ASC on gold aspects remains relatively stable, indicating that span extraction is the dominant bottleneck for end-to-end quality. Surprisingly, ASC accuracy on predicted aspects can exceed gold-aspect accuracy when the extractor returns a biased subset of "easier" spans; we interpret this pattern as a selection effect rather than genuine improvement. Complementing aggregate scores, we analyze row-normalized confusion matrices, ATE span-error distributions, and end-to-end error tracing that attributes mistakes to missed aspects, boundary errors, spurious terms, or ASC failures.

Introduction

Online reviews drive decisions in hospitality, retail, and technology markets. Stakeholders rarely need only a single score for an entire document; they need which aspects of a product or service were praised or criticized. Aspect-based sentiment analysis formalizes this need by linking targets (e.g., *battery*, *waitstaff*) to polarities (positive, negative, neutral). Industrial pipelines often decompose the problem: first detect spans that denote aspects, then classify sentiment conditioned on each span. Such modular designs reuse strong sequence taggers and sentence-pair classifiers, simplify debugging, and mirror production constraints where components may be maintained by different teams or updated on different cadences.

Modularity comes with a cost: errors compound. An extractor that misses a negative aspect entirely prevents the classifier from ever recovering that signal; an extractor that proposes a slightly wrong boundary may feed the classifier a misleading cue word; spurious aspects generate sentiment labels that do not exist in the annotation. Understanding where failures arise is therefore as important as reporting a single end-to-end F1. Prior work on ABSA frequently reports joint models or pipeline results, but cross-domain studies often emphasize sentiment or domain adaptation without isolating the ATE stage's contribution in a fully wired system.

This work documents an end-to-end ATE $\rightarrow$ ASC pipeline for the SemEval-2014 restaurant and laptop domains (Pontiki et al., 2014). We compare BERT-base-uncased and DeBERTa-base within a common preprocessing and evaluation pipeline, using model-specific hyperparameters while holding data splits consistent across teammates. The experimental matrix covers eight settings: for each backbone, we test in-domain (train and test on the same domain) and cross-domain (train ATE/ASC on one domain, evaluate on the other domain's test set) in both directions. We align evaluation with two ASC regimes: oracle aspects (gold spans), which upper-bound sentiment quality absent extraction noise, and predicted aspects, which reflect realistic pipeline behavior.

Our contributions are empirical and diagnostic: (i) a complete numeric dashboard for cascade ABSA on standard benchmarks with modern en-

coders; (ii) explicit error-propagation measurement between gold- and predicted-aspect ASC; and (iii) a structured error analysis—confusion matrices, span-level ATE error typing, and end-to-end attribution—grounded in exported evaluation artifacts. We discuss when improving ATE should take priority over improving ASC, particularly under domain shift, and when aggregate accuracy can be misleading due to extractor-induced selection bias.

Positioning. Neural joint ABSA systems (span + sentiment in one objective) can implicitly share representations; however, cascaded systems remain widespread because they map cleanly to existing tooling, allow independent iteration on tagging versus classification, and support cross-domain reuse of a sentiment head when only the extractor is retrained. Our study does not claim state-of-the-art joint modeling; instead, it quantifies cascade behavior under transparent assumptions that match many engineering roadmaps.

Task and Data

Task definition

We address two subtasks:

1. **Aspect term extraction (ATE).** Given a sentence, predict token spans corresponding to explicit aspect mentions. We treat this as a sequence-labeling problem with BIO tags over subword tokens and map predictions back to word-level spans for evaluation and for feeding ASC.

2. **Aspect sentiment classification (ASC).** Given a sentence and a single aspect string, predict one of three labels: positive, negative, or neutral. Following common practice for SemEval-2014, conflict labels are excluded from the three-way setup.

The end-to-end objective is to output a set of (aspect, sentiment) pairs aligned with test-time annotations.

Corpora

We use the SemEval-2014 Task 4 datasets for restaurants and laptops (Pontiki et al., 2014): official training data for model fitting, a fixed sentence-level train/validation split, and the official gold test sets for final evaluation. Restaurant and laptop reviews differ lexically and pragmatically: food discourse involves taste and service vocabulary; laptop discourse emphasizes hardware attributes (*keyboard*, *fan noise*) and performance. These differences motivate domain-conditioned models and stress-test cross-domain generalization when train and test domains diverge. Both domains use the same three-way polarity inventory for ASC, which keeps metrics comparable across experiments.

Preprocessing and splits. Training examples for ATE and ASC are tokenized consistently with the corresponding pretrained encoders and preserve aligned train/validation partitions at the sentence level. The test partition is always the official SemEval gold test set for the target domain. We do not perform additional random test splits; all numbers reported here are therefore comparable across backbones for a given experiment setting.

Metrics

ATE is scored with span-level precision, recall, and F1 against gold aspects on the test set. ASC is scored with accuracy and macro-averaged F1 over the three classes. For pipeline analysis we compute ASC twice: ASC (gold) feeds the classifier annotated aspects; ASC (pred) feeds it predicted aspects from ATE. The error-propagation gap is

$$\begin{aligned} \Delta_{\text{acc}} &= \text{Acc}_{\text{gold}} - \text{Acc}_{\text{pred}}, \\ \Delta_{\text{F1}} &= \text{Macro-F1}_{\text{gold}} - \text{Macro-F1}_{\text{pred}}. \end{aligned} \tag{1}$$

Positive Δ means predicted aspects hurt sentiment relative to oracle spans; negative Δ means predicted aspects appear easier than gold—typically when predictions omit difficult aspects or introduce spurious easy-to-classify terms.

End-to-end pairing. For each gold aspect, pipeline evaluation checks whether a matching predicted aspect (under the project's span-matching rules) carries the correct sentiment. Unmatched gold aspects count against recall; unmatched predicted aspects can create false positives if spurious spans receive confident sentiment predictions. This coupling is why ATE recall especially influences whether ASC errors are even measurable on full gold coverage.

System Description

Overview

The system cascades independently trained ATE and ASC modules. The ATE component emits, for each test sentence, a list of predicted aspect strings.

Each (sentence, aspect) pair is then passed to the ASC module, which returns a polarity label. Final end-to-end predictions are the union of predicted aspects with their predicted sentiments; evaluation uses standard matching against gold (aspect, sentiment) pairs.

Models and training

Backbones and hyperparameters are as follows:

- BERT: `bert-base-uncased` (Devlin et al., 2019) with 5 training epochs for both ATE and ASC, learning rate 3e-5, batch size 16 (train) / 32 (eval), and weight decay 0.01 where applicable.
- DeBERTa: `microsoft/deberta-base` (He et al., 2021) with ATE at batch size 8, learning rate 2e-5, and 5 epochs; ASC at batch size 16, learning rate 2e-5, and 3 epochs.

ATE is formulated as token classification, whereas ASC uses a sentence-pair encoding [CLS] sentence [SEP] aspect [SEP]. Domain-specific weights are trained separately for restaurant and laptop.

Why ASC is evaluated twice. The gold-aspect score answers: "How good is sentiment given perfect targets?" The predicted-aspect score answers: "What does a user actually get after extraction?" The gap between them isolates cascade loss attributable to span errors. In isolation, a high gold-aspect ASC with low predicted-aspect ASC points to ATE as the lever; when both are low, ASC may need better features or more data even with clean spans.

Cross-domain protocol

For each backbone, we evaluate eight experiments: (train domain, test domain) $\in \{\text{restaurant}, \text{laptop}\}^2$, covering in-domain baselines and both cross-domain directions. ATE and ASC weights always come from the same training domain; only the test domain changes. This isolates domain shift at evaluation time while keeping the training recipe fixed.

Analysis artifacts

Beyond scalar metrics, we analyze confusion matrices (absolute counts and row-normalized versions), ATE span-error statistics (missing, spurious, boundary errors), end-to-end error-tracing histograms, and representative qualitative examples.

ID	Train → Test	Backbone	Type
1	restaurant → restaurant	BERT	in-domain
2	restaurant → restaurant	DeBERTa	in-domain
3	restaurant → laptop	BERT	cross-domain
4	restaurant → laptop	DeBERTa	cross-domain
5	laptop → restaurant	BERT	cross-domain
6	laptop → restaurant	DeBERTa	cross-domain
7	laptop → laptop	BERT	in-domain
8	laptop → laptop	DeBERTa	in-domain

Experimental Setup

Table 1 lists the eight experimental settings considered in this study.

Implementation and hardware. Experiments were implemented in Python with PyTorch and Hugging Face Transformers. Training was conducted on NVIDIA GPUs; exact runtimes varied by model and domain.

Results

Aggregate pipeline metrics

Table 2 reports ATE and ASC metrics and propagation gaps.

Cross-domain ATE collapse. When training on restaurant and testing on laptop (IDs 3–4), ATE F1 falls to 0.25–0.32, far below in-domain restaurant (0.56–0.58). The reverse direction (laptop → restaurant, IDs 5–6) shows high precision but very low recall (e.g., 0.66 precision with 0.16 recall for BERT), yielding low F1 (0.26–0.36). These patterns indicate strong lexical and structural domain gaps for span identification.

ASC on gold aspects remains comparatively stable across conditions (accuracy often in the 0.76–0.84 band), suggesting that—with explicit targets—the sentiment model is not the primary failure mode under shift in this setup.

Error-propagation gaps. Several runs show negative Δ_{acc}: predicted-aspect accuracy exceeds gold-aspect accuracy (IDs 1, 3, 4, 5, 6, 7, 8 for accuracy). This arises when the extractor drops hard-to-classify aspects or focuses on salient sentiment cues, making the predicted subset easier than the full gold set. The strongest effect appears for ID 5 (laptop → restaurant, BERT): gold accuracy 0.8018 versus predicted 0.9218, with $\Delta_{\text{acc}} = -0.12$. For the same run, predicted-aspect macro-F1 also exceeds gold (0.8283 vs. 0.6899, $\Delta_{\text{m-F1}} = -0.1384$), showing that the selection effect can inflate not only headline accuracy but also a class-balanced metric.

ID	ATE P	ATE R	ATE F1	ASC Acc (gold)	ASC m-F1 (gold)	ASC Acc (pred)	ASC m-F1 (pred)	Δ Acc	Δ m-F1
1	0.5558	0.5658	0.5608	0.8304	0.7439	0.8483	0.7589	-0.0179	-0.0150
2	0.5691	0.5828	0.5759	0.8420	0.7481	0.8405	0.7288	0.0015	0.0193
3	0.3244	0.2098	0.2548	0.7618	0.7204	0.7852	0.7192	-0.0234	0.0012
4	0.4116	0.2681	0.3247	0.7947	0.7440	0.8198	0.7320	-0.0251	0.0120
5	0.6605	0.1603	0.2579	0.8018	0.6899	0.9218	0.8283	-0.1200	-0.1384
6	0.6228	0.2543	0.3611	0.7982	0.67	0.8838	0.6770	-0.0856	-0.0070
7	0.5097	0.5379	0.5234	0.7571	0.6971	0.7755	0.6895	-0.0184	0.0076
8	0.5452	0.5521	0.5486	0.7947	0.7547	0.8011	0.7436	-0.0064	0.0111

BERT versus DeBERTa

In-domain restaurant (1 vs. 2): DeBERTa improves ATE F1 (0.5759 vs. 0.5608) and gold ASC accuracy (0.842 vs. 0.8304). For predicted aspects, BERT achieves slightly higher ASC accuracy (0.8483 vs. 0.8405) and macro-F1 (0.7589 vs. 0.7288), highlighting a metric-dependent comparison between gold-aspect and predicted-aspect evaluation.

In-domain laptop (7 vs. 8): DeBERTa improves ATE F1 (0.5486 vs. 0.5234) and gold ASC metrics. On predicted aspects, DeBERTa leads in accuracy (0.8011 vs. 0.7755) and macro-F1 (0.7436 vs. 0.6895), with smaller negative accuracy gaps—consistent with better extraction reducing harmful mismatch.

Cross-domain: DeBERTa's ATE F1 exceeds BERT's for restaurant → laptop (0.3247 vs. 0.2548) and laptop → restaurant (0.3611 vs. 0.2579), showing encoder upgrades help span ID under shift, though F1 remains low in absolute terms.

Interpreting laptop → restaurant (IDs 5–6). The extractor's precision–recall asymmetry is extreme: many predicted spans are locally plausible (high precision) yet most gold aspects are missed (low recall). For sentiment, evaluating on predicted aspects therefore skips numerous gold pairs—especially subtle or rare mentions—while ASC may still classify the surviving subset with high accuracy. This dynamic explains the large negative Δ_{acc} in ID 5, where predicted-aspect evaluation also yields a higher macro-F1 than gold: the model is not universally "better" on predicted inputs; both metrics are inflated by coverage bias on an easier surviving subset.

Interpreting restaurant → laptop (IDs 3–4). Both backbones suffer low ATE F1, but DeBERTa raises F1 by 7 points absolute versus BERT. ASC on gold remains mid-to-high seventies to low eighties in accuracy, reinforcing that target identification—not polarity modeling in isolation—is the limiting factor when moving to laptop test vocabulary.

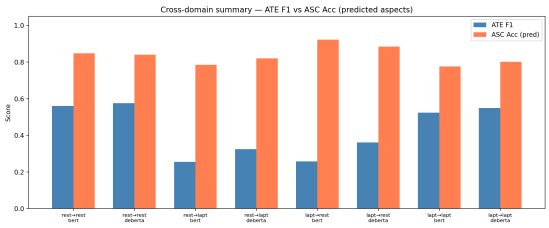

Figure 1: ATE F1 vs ASC accuracy on predicted aspects across eight experiments.

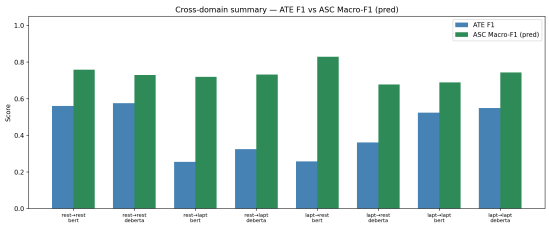

Figure 2: ATE F1 vs ASC macro-F1 on predicted aspects across eight experiments.

Summary figures

Figure 1 and Figure 2 plot ATE F1 against ASC accuracy / macro-F1 on predicted aspects for all eight experiments, summarizing the trade-off between extraction quality and downstream sentiment when aspects are predicted.

Qualitative and Error Analysis

In-domain restaurant (BERT): confusion, ATE errors, and E2E tracing

Figure 3 shows the row-normalized ASC confusion matrix when testing in-domain on restaurant data with BERT. Each row sums to one; off-diagonal mass in the neutral row highlights neutral-to-non-neutral confusion, a recurring issue under class imbalance and implicit or hedged sentiment. Reading this plot alongside Table 2 clarifies why macro-F1 should accompany accuracy: a classifier that under-predicts neutral can still achieve high accuracy while hurting minority-class recall.

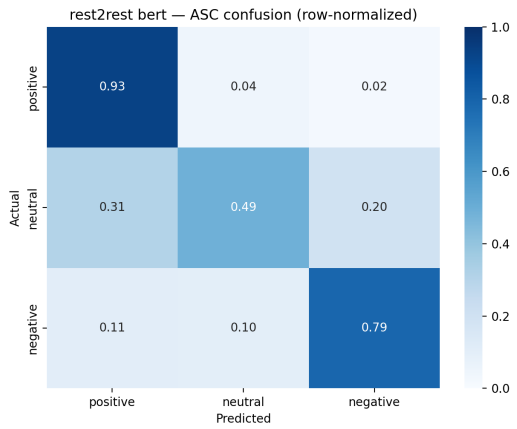

Figure 3: Row-normalized ASC confusion matrix — restaurant → restaurant, BERT (in-domain).

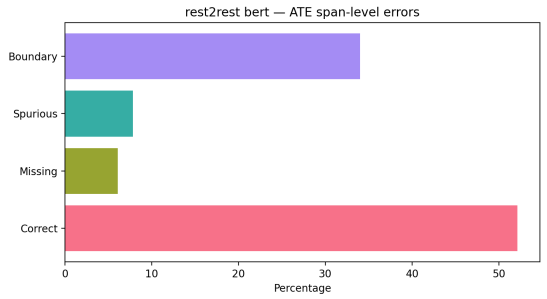

Figure 4: ATE span-level error distribution — restaurant → restaurant, BERT.

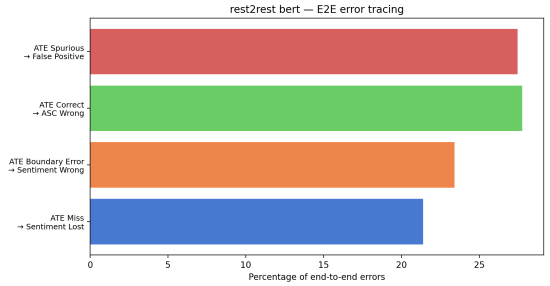

Figure 5: End-to-end error tracing — restaurant → restaurant, BERT.

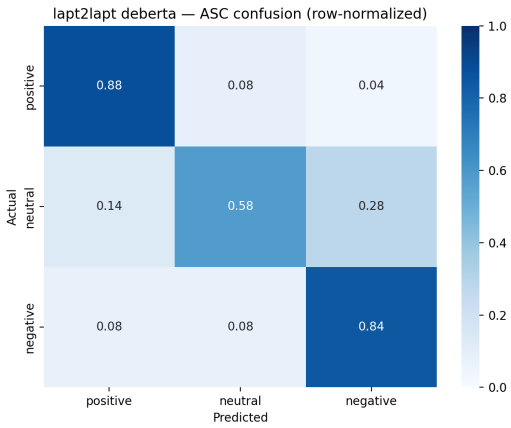

Figure 6: Row-normalized ASC confusion matrix — laptop → laptop, DeBERTa (in-domain).

Figure 4 breaks down ATE span-level errors (missing, spurious, boundary). Figure 5 attributes end-to-end failures to ATE miss, boundary mistakes, ASC errors, or spurious aspects. Together, Figures 3–5 give a coherent in-domain restaurant portrait before we compare domains.

In-domain laptop (DeBERTa): confusion, ATE errors, and E2E tracing

Figures 6–8 mirror the same three views for laptop test data with DeBERTa weights trained on laptop. Neutral–negative–positive confusions differ from restaurant (Figure 6), while ATE error shares (Figure 7) reflect laptop-specific multi-token aspects (e.g., compound product phrases). End-to-end tracing (Figure 8) shows how often ASC alone is responsible once spans are correct.

Cross-domain end-to-end tracing

Figures 9–10 focus on cross-domain settings. We categorize mistakes as ATE miss → sentiment lost, ATE boundary error → sentiment wrong, ATE correct → ASC wrong, and ATE spurious → false positive. Compared with in-domain Figures 5 and 8, missed aspects clearly gain relative mass when the extractor is trained on a mismatched domain lexicon, while boundary errors do not consistently increase in the two cases shown here.

Example error cases

Representative error cases illustrate concrete failure modes. For instance, an ATE miss on *people* in "I trust the people at Go Sushi…" removes the chance to classify that aspect's sentiment entirely (ATE Miss → Sentiment Lost). A boundary error on *sangria* versus *sangria's* can flip polarity when punctuation alters the substring seen by ASC (ATE Boundary Error → Sentiment Wrong). ASC-only errors include cases where the aspect string is correct but polarity is wrong (e.g., *staff* predicted positive where gold is negative), motivating contrastive or negation-aware training even when extraction is perfect.

Cross-domain qualitative themes (informal). When ATE trains on restaurant text and tests on laptops, we expect missed hardware attributes (*battery*, *keyboard backlight*) and spurious food-like tokens if the tagger hallucinates domain-inappropriate

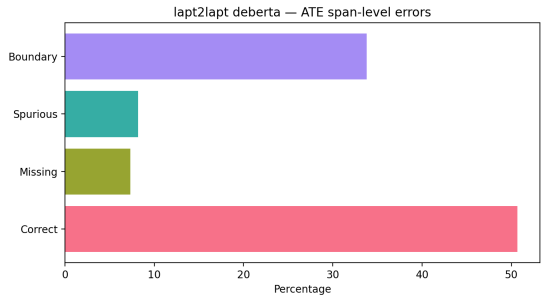

Figure 7: ATE span-level error distribution — laptop → laptop, DeBERTa.

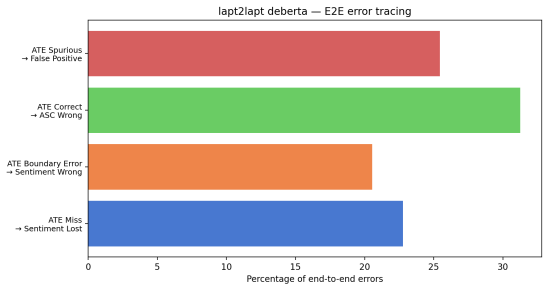

Figure 8: End-to-end error tracing — laptop → laptop, DeBERTa.

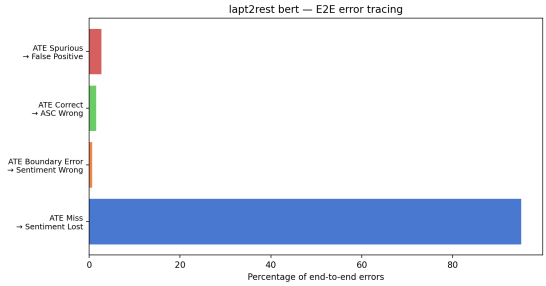

Figure 9: End-to-end error tracing — laptop → restaurant, BERT (cross-domain).

Figure 10: End-to-end error tracing — restaurant → laptop, DeBERTa (cross-domain).

spans. In the reverse direction, laptop-trained extractors may under-segment multi-word restaurant dish names. We do not manually quantify these patterns here; the error-tracing bars and ATE error-type histograms provide aggregate proxies, and future work could quantify these lexical effects more systematically.

Conclusion

We presented a cascaded ATE → ASC system for SemEval-2014 restaurant and laptop reviews with BERT and DeBERTa backbones, evaluated under eight train/test domain configurations. ATE quality—especially recall under cross-domain test—drops sharply and drives end-to-end risk, while ASC on gold aspects stays comparatively stable, indicating that span extraction should be prioritized when resources are limited. Propagation-gap analysis shows that accuracy on predicted aspects can exceed gold-aspect accuracy when extractors filter aspects to an easier subset; macro-F1 and pair-level metrics remain essential to avoid overstating sentiment quality. DeBERTa consistently helps ATE F1 in cross-domain pairs, but absolute F1 still calls for domain adaptation, data augmentation, or joint training. Future work includes joint modeling, calibrated spurious-mention control, and richer neutral-class handling informed by confusion matrices.

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